



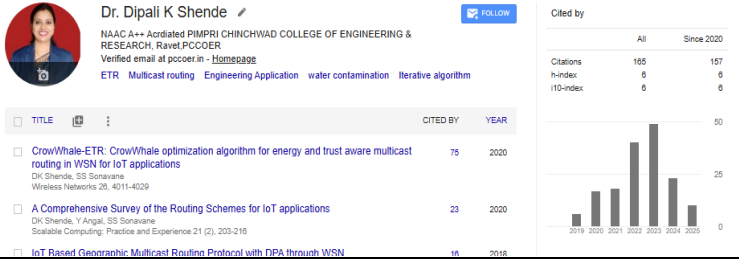
**Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering &
Research Ravet, Pune**



Academic Year:
2025 - 26

Faculty Profile

Name of Faculty:	Dr. Mrs. Dipali K. Shende		
Qualification :	Ph.D. [IoT], M.E.[VLSI/Embedded]		
Designation :	Associate Dean[Q & A] ,Assistant Professor		
Department :	Electronics and Telecommunication Engineering		
Date of Joining :	23/08/2022		
Experience :	Teaching: 19 Yrs Industry: -2 Yrs.		
Research Areas / Area of Specialization	Internet of Things, Embedded System, Wireless Sensor Network		
Courses / Subjects Taken:	▪ At PG - Embedded System Design, ESP ▪ At UG- IoT, IoT Honor, WSN, Advance Processor, Embedded Processor, RTOS, Data Structure, Embedded System Design, VLSI Design, Microprocessor Microcontroller and Application, System Programming and Operating System.		
IPR : Patent and Copyrights	Title	Docker No./ Diary No.	Registered /Filed
	Wearable device to monitor health of Neuro-Patients	6399147	Granted-UK Design October 2024
	Enhanced Wireless Connectivity Input Device for Visually Impaired Persons	202421080466	FER Phase , Indian Patent Journal dated 22 October 2024
	A Device for Rescue from Bore-Well	Patent Number- 202021022910	Granted-Indian 19 May 2022
	System to Alarm Freshwater Contamination Based on Intelligent Device with IoT-Based Multicast Routing	Patent Number: - 201921045885	Granted- Indian Patent January 2024
	A DEVICE FOR RESCUE FROM BORE-WELL	397178	Granted- Indian Patent December 2020
	Smart & Intelligent lamp pole consisting of Insect Killer Whether Station WIFI and EV charging station	Patent NO: - 202121059138	FER Phase- Indian Patent 23 Dec 2021
	An Electric Auto Toggle Holder	Patent NO: - 202121057198	FER Phase -12 Sept 2021

	Copyright on “Detection and simulation of pancreatic tumor growth based on prediction model”	Diary Number L-156628/2024	Registered –Nov 2024
	Copyright on “Whether Monitoring and Analysis Using CNN”	Diary Number L-137228-2023	Registered –Dec 2023
	Copyright -AI - Driven Crop Prediction and Soil Nutrient Estimation with NPK Analysis”	Diary Number L-137600-2023	Registered –Dec 2023
	Copyright Smart River Cleaning BoT	Diary Number L-122480/2023	Registered –May 2023
Profile Link :	📌 Google scholar URL: https://scholar.google.co.in/citations?hl=en&user=ONg2HpYAAAAJ 📌 Research-Gate URL:https://www.researchgate.net/profile/Dipali-Shende 📌 Vidwan ID : https://vidwan.inflibnet.ac.in/profile/414051 📌 Web of Science : https://www.webofscience.com/wos/author/record/JDM-4064-2023 📌 ORCID id: https://orcid.org/0000-0002-0238-7478		
Citation and H –Index:	Citation=160 and h-index=3,i-index-3 		
Awards/Appreciation /Achievements	▪ Awards=4 ▪ International=01 ▪ National=2 ▪ Institute Level=1		
Reviewer	▪ International Journal of Creative Research Thoughts IJCRT-13 sept 2017. ▪ Innovative Research in Engineering & Multidisciplinary Present Day Science [IJIRMPs] Since Aug 2018.		
Membership	IETE (M215190), IEEE [94428066]		
Resource Person /Expert Session/Jury Invited:	Workshops=15, Expert Lectures=18, FDP[Coordinated]=12,		
Workshop/FDP/ STTP	▪ Workshop=34 ▪ FDP =44		

attended(more than a Week):	STTP=06		
University level appointments:	Paper Setter for UG, PG –Subject RTOS since A.Y.2013-14 till date, PR/OR External Examiner, ESD- SPPU BE Subject Chairmen		
Administrative Duties:	- Ranking Incharge- NIRF Nodal Officer - Journal Editor - NAAC Scrutiny Committee Member - IETE Faculty In-charge - Value Addition Program [VAP] coordinator - WGC Department Member, Department Activity Monitor In-Charge, B.E. Project In charge, Prepared academic departmental report since A.Y. 2012 - 2015.		
Academics Duties:	Class Teacher and Subject In-charge for Data Structure and Embedded System, VLSI & Embedded Lab In-Charge		
Research Publications:	59-International [42], National [17]		
Journal Publications:	Year	Title of Paper	Published in
	Aug 24	An efficient fault diagnosis model using deep learning Model	Power Electronics- Inder science
	Aug - 23	Use of Generative Adversarial Network (GAN) under insufficient data	Springer Lecture Notes- Cognitive Computing Cyber Physical Systems
	Jan 2022	Sewage water management and healthcare monitoring in IoT using Optimized deep residual network	Published in SCI Indexed- Taylor and Francis journal - “The Journal of Experimental and Theoretical Artificial Intelligence” on 30 Oct 2022.
	March 2022	Performance analysis of the CrowWhale optimization with the existing optimizations for IoT multicast routing application	Web of Science and Scopus Indexed Journal “Web Intelligence” on 16 Dec 2022.
	2021	“An iterative Crow-Whale based optimization model for energy aware multicast routing in IoT”	Web of science Scopus Indexed Journal International Journal of Information Security and Privacy (IJISP), Volume 16, Issue 1, Article 13 DOI: 10.4018/IJISP.300317-
	2021	Aquaponics System for Modern India in UGC Approved Journal No: 63975	Journal of Emerging Technologies and Innovative Research (JETIR) , Vol. 8 Issue 05, May 2021 ISSN: 2349-5162.

	2020	A Comprehensive Survey of the Routing Schemes for IoT applications	Scopus Indexed Journal Scalable Computing: Practice and Experiences Volume 21 Issue 2 DOI 10.12694:/scpe.v21i2.1667-29 th June 2020
	2020	Crow Whale-ETR: Crow whale optimization algorithm for energy and trust aware multicast routing in WSN for IoT applications	Springer SCI Indexed Journal Wireless Networks- The Journal of Mobile DOI: https://doi.org/10.1007/s11276-020-02299-y
	2020	Bandicoot: Drain Clog Detection and Prevention to support Swatch Bharat Abhiyan	AEGAEUM JOURNAL-UGC Care Approved Group II Journal Vol. 8, Issue 06, ISSN: 0776-3808.
	2020	Detection of Water Contamination with Respect to Different Parameters Using IOT Based Multicast Routing	Journal of Scientific Computing Vol. 9, Issue 05, ISSN: 1524-2560.
	2020	"Embodiment Well Rescue Intelligent Device with Real Time Feed for Safety Lift"	AEGAEUM JOURNAL-UGC Care Approved Group II Journal Vol. 8, Issue 04,2020 ISSN: 0776-3808.
	2019	IOT Based Energy Meter Billing and Monitoring System"	International Journal for Scientific Research & Development IJSRD Vol. 7, Issue 03, 2019 ISSN: 2321-0613.
	2019	Nirbhaya" with U-A Women's Safety Device"	International Journal of Scientific Research and Engineering Development UGC Care Journal, Volume 2 Issue 3, 2019 ISSN : 2581-7175
	2019	Dementia Patient Activity Monitoring and Fall Detection using IOT for Elderly"	International Journal of Trend in Scientific Research and Development (IJTSRD) UGC Care Journal, Volume: 3 Issue: 4 May-Jun 2019 ISSN: 2456 – 6470.
	2018	Automation of dry-wet waste collection to support Swachh Bharat Abhiyan and its monitoring over IOT enabled WSN	International Journal of Computer Sciences and Engineering,UGC Care Journal,Vol.-6, Issue-6, June 2018 E-ISSN: 2347-2693
Notable Contributions		➤ Book published with Scientific International Publishing House,Title "Machine Learning" in 2025, ISBN No.- 978-93-6674-194-9. ➤ Book published titled Computer Based Optimization ISBN-978-93-7128-268-0 in 2025 ➤ Received appreciation as Reviewer from Scopus and Springer Nature Articles ➤ Session chair in IEEE ICITEICS 2024 International conference at VIT, Bengaluru ,28-29 June 2024. ➤ Syllabus Revision committee member for IoT subject in Autonomy. ➤ Editor for Journal of Electronic Design and computational Science PCCOER. ➤ Reviewer in Springer International conference ETMCIS 2024 at MAE	

	Alandi,13-15 June 2024. ➤ Completed Industry projects with Magna and Aashay Measurement under MOU. ➤ Coordinated International FDP in 2024 in collaboration with MathWorks India. ➤ Book Reviewer for Book Chapter under BP publication. ➤ Program Chair in International Conference at PCCOER, Ravet during 19-20 April 2024. ➤ Session chair in 3rd IEEE -ASIANCON 2023 International conference held at PCCOER, Ravet during Aug 2023. ➤ Reviewer in International conference on Green Energy held at AISSMSCOE, Pune during May 2023. ➤ Member for IoT Autonomy Syllabus Revision SE Subject Revision-SNJB Chandwad ➤ Reviewer in International conference on Green Energy held at AISSMSCOE, Pune during May 2023. ➤ Session coordinator at ASIANCON 2022 held at PCCOER, Ravet during Aug 2022. ➤ Avishkar State Level Selection of Research application in PPG category at Pune University ➤ Webinar conducted on “Application of IoT” at SIT, Lonavala on 19 Jan 2024. ➤ Delivered Expert Lecture in ATAL-AICTE FDP at JSPM-BSIOTR Wagholi, Pune on 20 Dec 2023- Topic Web of Things. ➤ Delivered Expert Lecture in faculty orientation workshop of BE subject Embedded System RTOS SPPU Pune, at SITRC Nashik during 30/01/2023 to 01/02/2023. ➤ Delivered Expert Lecture in National level faculty development workshop on Recent Trends in AI, IoT and Data Analytics during 3-8 Oct 2022 at PCCOER, Ravet. ➤ Coordinated one Week National Level Faculty Development program during 3-8 Oct 2022. ➤ Coordinated National level student Conference TKD21 at SIT Lonavala on 30th May 2021. ➤ Received reviewer certificate for review IoT track paper in National Conference TKD21 at SIT Lonavala on 30th May 2021. ➤ Completed in 4-week NPTEL Course subject-C & Assembly scored Silver Elite Certification 86%. ➤ Participated in 8-week NPTEL Course subject- Embedded System and ARM scored Silver Elite Certification and also received 1 week FDP from AICTE for same. Duration Jan 19 to Mar 19. ➤ Received Award of Best Event “Code Studio Dive into Python” in Technical festival Techtonic 2k19 held on 20-22 Feb 2019 at STES Institute level. ➤ Received QIP funding of 1,80000/- from SPPU in A.Y. 2018-19 for IOT. ➤ Participated and in two consecutive FDP101X and 201X conducted by IIT
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	MOOC Course at National Level and scored 97 % in Teaching Learning ICT Tools with Certification during Jul 2018 to Jan 2019. ➤ Awarded of Elite Certificate from NPTEL IIT Madras in Mar 2018 for two subject Introduction to Research and Internet of Things Dec 2018. ➤ Inaugurated “Center of Excellence IoT” in A.Y. 2018 in E & TC Dept in association with vInnovate Lonavala, and every year till date arranging workshop, innovation competitions, guest lectures to help students for enhancing skills. ➤ Received Appreciation Letter for IETE Forum In-charge at department level since 2014- Coordinated altogether 9 activities in 2019-21. ➤ Received Appreciation Letter from Telematics Delhi for organizing Internship program on 18Jul 17. ➤ Received Appreciation Letter from Industry Immediata Pune for arranging IV for TE students on 17 Jan 17. ➤ Received Appreciation Letter from Snehwan Orphanage, Pune for arranging student visit 2017. Appreciated for the Report Preparation of Application towards- Best department Award [A.Y.11-12,14-15 and 20-21].
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Dr. Mrs. Dipali K. Shende
Dept. E & TC
dipali.shende@pccoer.in